Convex Clustering and Mixture Modeling: Extensions via Structured Weighting, l_1 Fusion, and Deterministic Recovery Guarantees

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Abstract

We study convex formulations for clustering and mixture regression, motivated by the limitations of classical non-convex approaches such as Expectation-Maximization (EM), which are sensitive to initialization and prone to local minima. Convex clustering, based on pairwise fusion penalties, provides a principled alternative with strong optimization guarantees, but existing formulations are largely restricted to specific weighting schemes and symmetric data distributions.

In this work, we revisit convex clustering from multiple perspectives, including optimization, statistical recovery, and its connections to mixture models and compressed sensing. We consider a generalized convex clustering framework that incorporates flexible weighting strategies and investigate extensions such as ℓ_1-based fusion penalties and non-symmetric distributions.

Furthermore, we explore the role of concentration inequalities and multiplicative weight mechanisms in deriving sharper recovery guarantees for such models. Our approach leverages efficient convex optimization techniques and aims to bridge the gap between theoretical guarantees and practical applicability. Preliminary analysis suggests improved robustness and broader applicability compared to standard convex clustering formulations.

1. Introduction

Clustering and mixture modeling form the backbone of many modern machine learning systems, particularly in unsupervised learning and density estimation. Classical approaches such as k-means and Gaussian Mixture Models (GMMs) rely on non-convex optimization procedures, which are sensitive to initialization and often converge to local minima. While probabilistic formulations provide modeling flexibility, they lack global optimality guarantees.

This motivates the study of convex formulations for clustering, which aim to retain interpretability while ensuring well-posed optimization. In particular, we consider the sum-of-norms convex clustering formulation:

$$\min_{x_1,\ldots,x_n\in\mathbb{R}^d} \frac{1}{2}\sum_{i=1}^{n}\|x_i-a_i\|^2+\lambda\sum_{1\le i<j\le n}\|x_i-x_j\|. \quad (1)$$

Although this formulation admits strong theoretical guarantees under idealized assumptions such as spherical Gaussian mixtures and uniform weighting, its applicability to more general data distributions and realistic settings remains limited.

In this work, we investigate extensions of convex clustering beyond these restrictive assumptions. In particular, we study the effect of incorporating structured weights into the fusion term, replacing the uniform penalty with:

$$\sum_{i<j} w_{ij}\|x_i-x_j\|, \quad (2)$$

where the weights w_{ij} encode similarity between data points.

We further explore alternative fusion penalties, such as ℓ_1-based formulations, and analyze their impact on cluster recovery and sparsity of the solution. Additionally, we consider clustering under non-symmetric and non-Gaussian distributions, where classical concentration arguments no longer directly apply.

A key theoretical direction involves understanding recovery guarantees under more general probabilistic models, including mixtures governed by concentration inequalities beyond the Gaussian setting. In particular, we aim to characterize conditions under which convex clustering can successfully recover cluster structure when data exhibits heavytailed or discrete behavior, such as sums of Bernoulli random variables.

Finally, we investigate adaptive weighting schemes, including multiplicative weight updates, that do not rely on strong prior assumptions about the data distribution. The goal is to develop a more general and robust convex clustering framework that bridges the gap between theoretical guarantees and practical applicability. In addition, we explore the integration of ideas from compressed sensing and Gaussian Processes to incorporate structured sparsity and kernel-based similarity into the convex clustering framework.

1.1. Gaussian Mixture Models and Expectation-Maximization (EM) Algorithm

Clustering and mixture modeling form the backbone of many modern machine learning systems, particularly in unsupervised learning and density estimation. Classical approaches such as k-means and Gaussian Mixture Models (GMMs) rely on non-convex optimization procedures, which are sensitive to initialization and often converge to local minima. This motivates the study of convex formulations that provide stronger optimization guarantees while retaining modeling flexibility.

The multivariate Gaussian distribution is defined as:

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) \tag{3}$$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right) \tag{4}$$

Key properties:

- Marginals and conditionals remain Gaussian
- Closed-form inference
- Fully characterized by mean and covariance

Maximum likelihood estimates:

$$\boldsymbol{\mu}_{ML} = \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n \tag{5}$$

$$\boldsymbol{\Sigma}_{ML} = \frac{1}{N}\sum_{n=1}^{N}(\mathbf{x}_n-\boldsymbol{\mu})(\mathbf{x}_n-\boldsymbol{\mu})^T \tag{6}$$

1.1.1. Gaussian Mixture Models

To model multi-modal data:

$$p(\mathbf{x}) = \sum_{k=1}^{K}\pi_k\,\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{7}$$

Latent variable formulation:

$$p(\mathbf{x}, z) = p(z)p(\mathbf{x} \mid z) \tag{8}$$

1.1.2. Expectation-Maximization (EM)

The log-likelihood is:

$$\log p(X) = \sum_{n=1}^{N}\log\left(\sum_{k=1}^{K}\pi_k\mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\right) \tag{9}$$

E-step:

$$\gamma(z_{nk}) = \frac{\pi_k\mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j\mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} \tag{10}$$

M-step:

$$N_k = \sum_n \gamma(z_{nk}) \tag{11}$$

$$\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_n \gamma(z_{nk})\mathbf{x}_n \tag{12}$$

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k}\sum_n \gamma(z_{nk})(\mathbf{x}_n-\boldsymbol{\mu}_k)(\mathbf{x}_n-\boldsymbol{\mu}_k)^T \tag{13}$$

$$\pi_k = \frac{N_k}{N} \tag{14}$$

1.1.3. Limitations

- Sensitive to initialization
- Converges to local optima
- No global optimality guarantees

1.2. Recovery of Mixture of Gaussians via Convex Clustering

We consider the convex clustering formulation known as *sum-of-norms clustering*. Given data points $a_1, \ldots, a_n \in \mathbb{R}^d$, the clustering problem is defined as:

$$\min_{x_1,\ldots,x_n \in \mathbb{R}^d} \frac{1}{2}\sum_{i=1}^{n}\|x_i - a_i\|^2 + \lambda \sum_{1\le i<j\le n}\|x_i - x_j\|. \tag{15}$$

This is a strongly convex optimization problem and admits a unique minimizer $x_1^*, \ldots, x_n^*$. The clustering structure is induced by the solution: points i and j belong to the same cluster if and only if $x_i^* = x_j^*$.

The parameter λ governs the clustering behavior. When $\lambda = 0$, each point forms its own cluster, while for sufficiently large λ, all points merge into a single cluster. As λ increases, clusters merge progressively, yielding a hierarchical structure.

We assume that the data is generated from a mixture of K spherical Gaussian distributions with means $\mu_1, \ldots, \mu_K$, variances $\sigma_1^2, \ldots, \sigma_K^2$, and mixing weights $w_1, \ldots, w_K$.

A central tool for analysis is the following characterization of clusters.

Theorem 1 (Cluster Characterization) Let x^* be the minimizer of the convex clustering problem. A subset $C \subseteq \{1, \ldots, n\}$ forms a cluster (i.e., $x_i^* = \hat{x}$ for all $i \in C$) if and only if there exist vectors z_{ij}^* for all $i, j \in C$, $i \neq j$, such that:

$$a_i - \frac{1}{|C|} \sum_{l \in C} a_l = \lambda \sum_{j \in C \setminus \{i\}} z_{ij}^*, \quad \forall i \in C, \tag{16}$$

$$\|z_{ij}^*\| \leq 1, \tag{17}$$

$$z_{ij}^* = -z_{ji}^*. \tag{18}$$

This provides necessary and sufficient conditions for cluster formation and enables local analysis of clusters.

The solution path satisfies a monotonic merging property.

Corollary 1 (Agglomeration Property) Let $\lambda' \geq \lambda$, and let $x^*(\lambda)$ and $x^*(\lambda')$ denote the corresponding solutions. If a set C forms a cluster at λ, then C remains contained within a cluster at λ'.

Thus, clusters merge as λ increases and never split, giving rise to a hierarchical clustering structure.

To analyze recovery, define the set of points close to each mean:

$$V_m = \{i \; : \; \|a_i - \mu_m\| \leq \theta\sigma_m\}, \quad m = 1, \ldots, K. \tag{19}$$

Theorem 2 (Recovery of Mixture of Gaussians) Fix $\theta > 0$ and $\epsilon > 0$. Then, with probability approaching 1 exponentially fast as $n \to \infty$, the following holds:

- All points in V_m are assigned to the same cluster if

$$\lambda \geq \frac{2\theta\sigma_m}{(F(\theta, d)w_m - \epsilon)n}. \tag{20}$$

- Clusters corresponding to different components $m \neq m'$ remain distinct if

$$\lambda < \frac{\|\mu_m - \mu_{m'}\|}{2(n-1)}. \tag{21}$$

Here, $F(\theta, d)$ denotes the cumulative distribution function of a chi distribution with d degrees of freedom.

This result shows that convex clustering can recover the underlying mixture structure without requiring a strict separation gap between clusters. Notably, the separation condition does not depend on the sample size n, unlike earlier approaches.

The key idea is that recovery is guaranteed for points within a fixed radius around each cluster center. This is essential since, in high-sample regimes, points from different components may become arbitrarily close.

Despite these strong guarantees, certain limitations remain. The analysis assumes spherical Gaussian distributions and uniform weighting in the fusion term. Moreover, recovery is only ensured for points sufficiently close to the true means.

These limitations motivate the study of more general convex clustering formulations, including alternative weighting schemes, non-symmetric distributions, and different fusion penalties, with the aim of extending recovery guarantees to broader settings.

2. Related Works

2.1. Gaussian Processes

Gaussian Processes (GPs) provide a non-parametric Bayesian framework for modeling distributions over functions. Unlike parametric models that assume a fixed functional form, Gaussian Processes define a prior directly over functions, allowing flexible modeling of complex data.

A Gaussian Process is defined as a collection of random variables such that any finite subset follows a multivariate Gaussian distribution. Formally, a function $f(\mathbf{x})$ is said to follow a Gaussian Process if:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \tag{22}$$

where $m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$ is the mean function and $k(\mathbf{x}, \mathbf{x}')$ is the covariance (kernel) function.

Given a set of inputs $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, the function values follow:

$$\mathbf{f} = [f(\mathbf{x}_1), \ldots, f(\mathbf{x}_n)]^T \sim \mathcal{N}(\mathbf{m}, K), \tag{23}$$

where $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$.

A key component of Gaussian Processes is the kernel function, which encodes similarity between data points. Common choices include:

- Squared Exponential (RBF):

$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2}\right) \tag{24}$$

- Linear Kernel:

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{x}' \tag{25}$$

The kernel implicitly defines a geometry in the data space, which plays a crucial role in clustering and similarity-based methods.

In regression settings, given noisy observations $y_i = f(\mathbf{x}_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, the posterior distribution at a test point $\mathbf{x}_*$ is given by:

$$\mathbb{E}[f_*] = k_*^T (K + \sigma_n^2 I)^{-1} \mathbf{y}, \tag{26}$$

$$\mathrm{Var}(f_*) = k(\mathbf{x}_*, \mathbf{x}_*) - k_*^T (K + \sigma_n^2 I)^{-1} k_*, \tag{27}$$

where $k_* = [k(\mathbf{x}_*, \mathbf{x}_1), \ldots, k(\mathbf{x}_*, \mathbf{x}_n)]^T$.

From an optimization perspective, Gaussian Process inference involves solving linear systems of the form:

$$(K + \sigma_n^2 I)^{-1}\mathbf{y}, \tag{28}$$

which connects GP models to kernel methods and convex optimization.

The relevance of Gaussian Processes to clustering arises through the kernel function, which induces a similarity structure among data points. This perspective aligns with convex clustering formulations, where pairwise relationships between points play a central role. In particular, kernel-based similarities can be incorporated into weighted fusion penalties, providing a bridge between probabilistic models and convex optimization-based clustering approaches.

2.2. Mathematical Theory of Compressed Sensing

The Shannon-Nyquist theorem asserts that to reconstruct a signal f with bandwidth B, the sampling rate must be at least $2B$. In a finite-dimensional context, for a trigonometric polynomial

$$f(t) = \sum_{k=-M}^{M} x_k e^{2\pi i k t},$$

reconstruction is impossible if the number of samples $m < N = 2M + 1$. Compressed Sensing (CS) circumvents this by solving

$$Ax = y,$$

where $A \in \mathbb{C}^{m \times N}$ and $m \ll N$, provided the signal x is s-sparse (i.e., $\|x\|_0 \leq s$) or compressible.

For compressible vectors, the error is bounded by the ℓ_p-error of the best s-term approximation:

$$\sigma_s(x)_p := \inf\{\|x - z\|_p : z \text{ is } s\text{-sparse}\}.$$

2.2.1. Complexity and Convex Relaxation

Direct recovery via ℓ_0-minimization is NP-hard. This can be shown by reduction from the Exact Cover by 3-Sets problem to the noise-aware ℓ_0 problem:

$$\min \|z\|_0 \quad \text{subject to} \quad \|Az - y\|_2 \leq \eta.$$

Given 3-element subsets $\{C_i\}_{i \in [N]}$ of $[m]$, construct $y = [1, \ldots, 1]^T$ and define A such that

$$(a_i)_j = \begin{cases} 1 & \text{if } j \in C_i, \\ 0 & \text{otherwise.} \end{cases}$$

If a solution z exists with $\|z\|_0 = m/3$, it corresponds to an exact cover.

To maintain tractability, we use Basis Pursuit (BP):

$$x^\# = \arg\min_{z \in \mathbb{C}^N} \|z\|_1 \text{subject to} \quad Az = y \text{ or } \|Az - y\|_2 \leq \eta.$$

2.2.2. Recovery Guarantees and Matrix Properties

Success of BP is governed by the **Null Space Property (NSP)**: matrix A satisfies NSP of order s if for all $v \in \ker(A) \setminus \{0\}$,

$$\|v_S\|_1 < \|v_{\overline{S}}\|_1 \quad \text{for all card}(S) \leq s.$$

For stability against noise e where $\|e\| \leq \eta$, we require the **Robust Null Space Property**:

$$\|v_S\|_1 \leq \rho\|v_{\overline{S}}\|_1 + \tau\|Av\|, \quad \text{for } 0 < \rho < 1,\ \tau > 0.$$

Under this condition, the ℓ_1 error is bounded by

$$\|x - x^\#\|_1 \leq \frac{2(1+\rho)}{1-\rho}\sigma_s(x)_1 + \frac{4\tau}{1-\rho}\eta.$$

The **Restricted Isometry Property (RIP)** provides a stronger condition. The s-th restricted isometry constant δ_s is the smallest δ such that

$$(1-\delta)\|x\|_2^2 \leq \|Ax\|_2^2 \leq (1+\delta)\|x\|_2^2$$

holds for all s-sparse vectors x.

If $\delta_{2s} < \frac{4}{\sqrt{41}} \approx 0.6246$, then A satisfies the ℓ_2-robust NSP.

Coherence is defined as

$$\mu(A) := \max_{i \neq j} |\langle a_i, a_j \rangle|.$$

It relates to RIP via

$$\delta_s \leq \mu_1(s-1) \leq (s-1)\mu.$$

The Welch bound provides a lower bound on coherence:

$$\mu \geq \sqrt{\frac{N-m}{m(N-1)}}.$$

2.2.3. Greedy Algorithms and Random Matrices

Alternative recovery methods include greedy algorithms such as Orthogonal Matching Pursuit (OMP) and CoSaMP.

OMP iteratively selects indices according to

$$j_{n+1} = \arg\max_j |(A^*(y - Ax^n))_j|\,.$$

Exact recovery for OMP after s iterations is guaranteed if

$$\mu_1(s) + \mu_1(s-1) < 1.$$

This step is computationally expensive.

CoSaMP uses a hard thresholding operator $H_s(z)$ that keeps the s largest entries. It requires the RIP condition

$$\delta_{13s} < \frac{1}{6}$$

for convergence.

For Gaussian or subgaussian matrices (with independent mean-zero entries of variance 1), the RIP condition $\delta_s \leq \delta$ holds with high probability provided

$$m \geq C\delta^{-2}\left(s\ln\left(\frac{eN}{s}\right)+\ln\left(\frac{2}{\epsilon}\right)\right).$$

In non-uniform recovery, a fixed s-sparse vector x can be recovered with high probability if

$$m > 2s\ln\left(\frac{2N}{\epsilon}\right).$$

A practical realization is the single-pixel camera, where an image $z = Wx$ is measured as

$$y = AWx,$$

using a single photodiode and random measurement patterns.

The connection between compressed sensing and convex clustering arises through the role of sparsity-inducing regularization. In compressed sensing, the use of ℓ_1-based penalties promotes sparse solutions, enabling recovery of high-dimensional signals from limited observations. A similar principle can be leveraged in convex clustering by encouraging sparsity in pairwise differences between variables.

In particular, replacing or augmenting the fusion penalty with sparsity-inducing norms can promote structured clustering, where only a subset of pairwise relationships actively contributes to cluster formation. This is analogous to selecting a sparse set of significant coefficients in compressed sensing. Furthermore, theoretical tools such as concentration inequalities and recovery guarantees from compressed sensing provide a framework for analyzing when such convex formulations can reliably recover underlying structure from noisy or high-dimensional data.

From this perspective, convex clustering can be viewed as a structured recovery problem, where the goal is not only to approximate the data but also to recover an underlying low-complexity representation in terms of clusters. This suggests that techniques and guarantees from compressed sensing can be adapted to study stability, robustness, and sample complexity in clustering formulations.

3. Extension: Sum-of-Norms Clustering with ℓ_1 Norm

3.1. Problem Formulation

We consider the following convex clustering problem with ℓ_1 norms:

$$\min_{x_1,\dots,x_n\in\mathbb{R}^d} f(x) = \frac{1}{2}\sum_{i=1}^{n}\|x_i-a_i\|_1+\lambda\sum_{1\leq i<j\leq n}\|x_i-x_j\|_1. \tag{29}$$

Since the objective is convex, any minimizer x^* satisfies:

$$0 \in \partial f(x^*). \tag{30}$$

3.2. Subgradient Characterization

For any vector $u \in \mathbb{R}^d$, the subdifferential of the ℓ_1 norm is:

$$\partial\|u\|_1 = \left\{ s\in\mathbb{R}^d : s_k = \begin{cases}\text{sign}(u_k), & u_k \neq 0,\\ \text{any value } \in[-1,1], & u_k = 0\end{cases}\right\}. \tag{31}$$

Thus any $s \in \partial\|u\|_1$ satisfies $\|s\|_\infty \leq 1$.

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3.3. Optimality Conditions

Let $x^* = (x_1^*,\dots,x_n^*)$ be a minimizer. Then for each i:

$$0 = \frac{1}{2}g_i + \lambda\sum_{j\neq i} w_{ij}, \quad \forall i, \tag{32}$$

where

$$g_i \in \partial\|x_i^* - a_i\|_1, \quad w_{ij} \in \partial\|x_i^* - x_j^*\|_1,$$

with

$$w_{ij} = -w_{ji}, \quad \|w_{ij}\|_\infty \leq 1.$$

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3.4. Cluster Characterization Theorem

Let x^* be an optimizer of (29). Let $C \subset \{1,\dots,n\}$.

(a) Necessary condition: Suppose there exists $\hat{x}$ such that

$$x_i^* = \hat{x}, \quad \forall i\in C, \quad x_i^* \neq \hat{x}, \quad \forall i \notin C.$$

Then there exist vectors z_{ij}^* for $i,j \in C$, $i \neq j$, satisfying:

$$g_i - \frac{1}{|C|}\sum_{l\in C} g_l = 2\lambda\sum_{j\in C\setminus\{i\}} z_{ij}^*, \quad \forall i \in C, \tag{33}$$

$$\|z_{ij}^*\|_\infty \leq 1, \tag{34}$$

$$z_{ij}^* = -z_{ji}^*. \tag{35}$$

(b) Sufficient condition: If such z_{ij}^* exist, then there exists $\hat{x}$ such that

$$x_i^* = \hat{x}, \quad \forall i \in C.$$

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3.5. Proof of Necessity

Since x^* minimizes f, we use (32).

For any $i \in C$, using $x_i^* = \hat{x}$:

$$0 = \frac{1}{2}g_i + \lambda\sum_{j\notin C} w_{ij} + \lambda\sum_{j\in C\setminus\{i\}} z_{ij}, \tag{36}$$

where $z_{ij} := w_{ij}$ for $i,j \in C$.

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Step 1: Summation over C Summing (36) over $i \in C$:

$$\frac{1}{2}\sum_{i\in C} g_i + \lambda \sum_{i\in C}\sum_{j\notin C} w_{ij} + \lambda \sum_{i\in C}\sum_{j\in C\setminus\{i\}} z_{ij} = 0. \quad (37)$$

Using antisymmetry:

$$\sum_{i\in C}\sum_{j\in C\setminus\{i\}} z_{ij} = 0,$$

we obtain:

$$\frac{1}{2}\sum_{i\in C} g_i + \lambda \sum_{i\in C}\sum_{j\notin C} w_{ij} = 0. \quad (38)$$

—

Step 2: Averaging Assuming w_{ij} is identical for all $i \in C$ for a fixed $j \notin C$ (since $x_i^* = \hat{x}$ for all $i \in C$), we have $\sum_{i\in C} w_{ij} = |C| w_{ij}$. Dividing (38) by $|C|$, we get:

$$\frac{1}{2|C|}\sum_{i\in C} g_i = -\lambda \sum_{j\notin C} w_{ij}. \quad (39)$$

—

Step 3: Subtraction Substitute $-\lambda \sum_{j\notin C} w_{ij}$ from (39) back into (36):

$$0 = \frac{1}{2} g_i - \frac{1}{2|C|}\sum_{l\in C} g_l + \lambda \sum_{j\in C\setminus\{i\}} z_{ij}. \quad (40)$$

Multiplying the entire equation by 2 and rearranging yields:

$$g_i - \frac{1}{|C|}\sum_{l\in C} g_l = -2\lambda \sum_{j\in C\setminus\{i\}} z_{ij}. \quad (41)$$

Since z_{ij} is antisymmetric ($z_{ij} = -z_{ji}$), we can absorb the negative sign, which proves (33).

□

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3.6. Proof of Sufficiency

We construct a feasible solution satisfying KKT conditions.

Define an auxiliary problem where all points in C share a single variable x, while others remain independent.

Let $(\tilde{x}, \tilde{x}_j)$ be its minimizer.

Define subgradients:

$$w_{ij} = \begin{cases} z_{ij}^*, & i, j \in C, \\ \text{valid } \ell_1 \text{ subgradient}, & \text{otherwise.} \end{cases}$$

Using the assumed conditions, one verifies that (32) holds for all i.

Hence the constructed point is optimal, implying:

$$x_i^* = \tilde{x}, \quad \forall i \in C.$$

□

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3.7. Discussion

Unlike the ℓ_2 case, the ℓ_1 formulation induces a polyhedral geometry due to the ℓ_∞ dual norm constraint.

4. Multiplicative Weighted Convex Clustering

Classical sum-of-norms clustering assigns uniform weights to all data points and pairwise interactions. While this setting enables recovery guarantees under idealized assumptions (e.g., spherical and well-separated Gaussian mixtures), it lacks robustness in heterogeneous or noisy environments.

To address this limitation, we introduce data-dependent weights. Let $r_i > 0$ denote the weight associated with data point $a_i \in \mathbb{R}^d$. The multiplicative weighted convex clustering problem is defined as:

$$\begin{aligned} F(x) = \min_{x_1,\ldots,x_n\in\mathbb{R}^d} & \frac{1}{2}\sum_{i=1}^{n} r_i |x_i - a_i|^2 \\ & + \lambda \sum_{1\le i<j\le n} r_i r_j |x_i - x_j|. \end{aligned} \quad (42)$$

The multiplicative structure $r_i r_j$ is essential: it preserves symmetry in the subgradient system and enables cancellation in pairwise sums. In contrast, general weights w_{ij} break this structure, preventing reduction to centroid-based characterizations and obstructing recovery analysis.

4.1. First-Order Optimality Conditions

The objective $F(x)$ is strictly convex and coercive, hence admits a unique minimizer $x^* = (x_1^*, \ldots, x_n^*)$. Optimality is characterized by the subgradient condition

$$0 \in \partial F(x^*).$$

For each $i \in 1, \ldots, n$, the gradient of the quadratic term is

$$\nabla_{x_i}\left(\frac{1}{2} r_i |x_i - a_i|^2\right) = r_i(x_i - a_i), \quad (43)$$

and the subdifferential of the fusion term is

$$\partial_{x_i}\left(\lambda \sum_{j\ne i} r_i r_j |x_i - x_j|\right) = \lambda \sum_{j\ne i} r_i r_j z_{ij}, \quad (44)$$

where $z_{ij} \in \partial|x_i - x_j|$ satisfies

$$z_{ij} = \begin{cases} \dfrac{x_i - x_j}{|x_i - x_j|}, & x_i \ne x_j, \\ |z_{ij}| \le 1, & x_i = x_j, \end{cases} \quad \text{and} \quad z_{ij} = -z_{ji}. \quad (45)$$

Combining the two terms, the first-order optimality condition becomes

$$r_i(x_i^* - a_i) + \lambda \sum_{j\ne i} r_i r_j z_{ij} = 0.$$

Dividing by $r_i > 0$, we obtain the KKT system

$$x_i^* - a_i + \lambda \sum_{j \neq i} r_j z_{ij} = 0, \quad \forall i \in 1, \ldots, n. \tag{46}$$

Thus, in the weighted formulation, each interaction term is scaled by the weight of the neighboring point, leading to an asymmetric but structured coupling across variables.

4.2. Cluster Characterization

Let $C \subseteq \{1, \ldots, n\}$ denote a subset of indices. Define the weighted mass and weighted centroid of C as

$$R_C = \sum_{l \in C} r_l, \qquad \bar{a}_C = \frac{1}{R_C} \sum_{l \in C} r_l a_l.$$

We characterize when all points in C collapse to a common value under the optimal solution. Specifically, C forms a cluster (i.e., $x_i^* = \hat{x}$ for all $i \in C$) if and only if there exist vectors z_{ij} such that

$$a_i - \bar{a}_C = \lambda \sum_{j \in C \setminus \{i\}} r_j z_{ij}, \quad \forall i \in C, \tag{47}$$

subject to

$$|z_{ij}| \leq 1, \qquad z_{ij} = -z_{ji}.$$

The multiplicative structure of the weights is essential in this characterization. Indeed, due to antisymmetry,

$$r_i r_j z_{ij} + r_j r_i z_{ji} = 0,$$

so all pairwise interaction terms cancel when summed over the cluster. This cancellation enables the reduction of the optimality system to a centroid-based form, which would not be possible under arbitrary pairwise weights.

4.3. Dual Certificate Construction

To verify that a given set V_m forms a cluster, we explicitly construct a feasible dual certificate. Let

$$R_{V_m} = \sum_{l \in V_m} r_l$$

denote the total weight of the set. Define

$$z_{ij}^* = \frac{1}{\lambda R_{V_m}} (a_i - a_j), \quad i, j \in V_m. \tag{48}$$

We now check that this choice satisfies the cluster condition. For any $i \in V_m$,

$$\lambda \sum_{j \in V_m \setminus \{i\}} r_j z_{ij}^* = \lambda \sum_{j \in V_m} r_j \left(\frac{a_i - a_j}{\lambda R_{V_m}} \right) \tag{49}$$

$$= \frac{1}{R_{V_m}} \left(a_i \sum_{j \in V_m} r_j - \sum_{j \in V_m} r_j a_j \right) \tag{50}$$

$$= a_i - \bar{a}_{V_m}. \tag{51}$$

Thus, the equality condition is satisfied. It remains to verify the norm constraint, which we address next.

4.4. Norm Bound Condition

For feasibility, the constructed dual variables must satisfy $|z_{ij}^*| \leq 1$. This requires

$$\frac{|a_i - a_j|}{\lambda R_{V_m}} \leq 1, \quad \text{for all } i, j \in V_m, \tag{52}$$

or equivalently,

$$|a_i - a_j| \leq \lambda R_{V_m}. \tag{53}$$

If all points in V_m lie within a ball of radius $\theta \sigma_m$ around their mean, then by the triangle inequality,

$$|a_i - a_j| \leq 2\theta\sigma_m.$$

Hence, a sufficient condition for feasibility is

$$\lambda \geq \frac{2\theta\sigma_m}{R_{V_m}}. \tag{54}$$

4.5. Lower Bound on λ

To express the bound in terms of observable quantities, we approximate the weighted mass as

$$R_{V_m} \approx |V_m| \mu_{r_m},$$

where μ_{r_m} denotes the mean weight within the component. Substituting this into the previous inequality yields

$$\lambda \geq \frac{2\theta\sigma_m}{|V_m| \mu_{r_m}}. \tag{55}$$

Under standard sampling assumptions for mixture models, the set size satisfies

$$|V_m| \approx (F(\theta, d) w_m - \epsilon) n,$$

which gives the explicit lower bound

$$\lambda \geq \frac{2\theta\sigma_m}{(F(\theta, d) w_m - \epsilon) n \mu_{r_m}}. \tag{56}$$

4.6. Upper Bound on λ (Cluster Separation)

To ensure that distinct clusters remain separated, we derive an upper bound on λ. From the KKT condition,

$$|x_i^* - a_i| \leq \lambda \sum_{j \neq i} r_j. \tag{57}$$

Consider $i \in V_m$ and $i' \in V_{m'}$ with $m \neq m'$. Using the triangle inequality,

$$|x_i^* - x_{i'}^*| \geq |\mu_m - \mu_{m'}| |x_i^* - a_i| |x_{i'}^* - a_{i'}| \tag{58}$$

$$\geq |\mu_m - \mu_{m'}| \lambda \sum_{j \neq i} r_j \lambda \sum_{j \neq i'} r_j. \tag{59}$$

Assuming concentration of weights so that

$$\sum_{j\neq i} r_j \approx (n-1)\bar{r},$$

we obtain the sufficient condition

$$\lambda < \frac{|\mu_m - \mu_{m'}|}{2(n-1)\bar{r}}. \tag{60}$$

4.7. Agglomeration Property

The multiplicative weighted formulation preserves the agglomerative structure of convex clustering. In particular, if a set C forms a cluster at some λ, then it remains fused for all $\lambda' \geq \lambda$. This follows from convexity of the objective and monotonicity of the subgradient feasibility conditions with respect to λ.

4.8. Summary of Changes

Original Formulation	**Weighted Formulation**
centroid	weighted centroid
$\lvert C\rvert$	$R_C = \sum r_i$
$\sum z_{ij}$	$\sum r_j z_{ij}$
cluster size	weighted mass
n	$n\bar{r}$

4.9. Key Insight

The success of this extension relies critically on the multiplicative structure $r_i r_j$. After differentiation, this structure reduces to a single factor r_j in the optimality system, preserving the symmetry required for pairwise cancellation. In contrast, arbitrary weights w_{ij} break this structure and prevent reduction to centroid-based characterizations, thereby obstructing the recovery analysis.

5. Numerical Experiments and Algorithmic Implementation

To validate the theoretical recovery guarantees of convex clustering and investigate its empirical performance across different data regimes, we implemented a custom optimization solver and conducted a series of computational experiments. The implementation strictly mirrors the theoretical bounds and data generation processes established in the literature.

5.1. ADMM Optimization Framework

The sum-of-norms convex clustering objective is non-differentiable due to the ℓ_2-norm fusion penalties. To optimize this efficiently, we employ the Alternating Direction Method of Multipliers (ADMM). The problem is formulated as:

$$\min_X \frac{1}{2}\sum_{i=1}^{n} \|x_i - a_i\|^2 + \lambda \sum_{i<j} w_{ij}\|x_i - x_j\|_2. \tag{61}$$

We introduce auxiliary variables $Z_{ij} \in \mathbb{R}^d$ such that $Z_{ij} = x_i - x_j$. The augmented Lagrangian with scaled dual variables U_{ij} and penalty parameter $\rho > 0$ yields the following iterative updates:

1. Z**-Update (Soft-Thresholding):** The auxiliary variables are updated via block soft-thresholding on the current differences augmented by the dual variables:

$$V_{ij} = x_i - x_j + U_{ij} \tag{62}$$

$$Z_{ij} = \max\left(1 - \frac{\lambda w_{ij}}{\rho\|V_{ij}\|_2}, 0\right) V_{ij}. \tag{63}$$

   Antisymmetry is explicitly enforced: $Z_{ij} = \frac{1}{2}(Z_{ij} - Z_{ji})$.
2. X**-Update (Primal Variables):** The update for x_i reduces to a strictly convex quadratic problem, which admits a closed-form solution:

$$x_i = \frac{a_i + \rho\sum_{j=1}^{n}(x_j + Z_{ij} - U_{ij})}{1+\rho n}. \tag{64}$$

3. U**-Update (Dual Variables):**

$$U_{ij} = U_{ij} + x_i - x_j - Z_{ij}. \tag{65}$$

The algorithm executes until both primal and dual residuals fall below a tolerance threshold of 10^{-6}. Continuous solutions x_i^* are then discretized into hard clusters by grouping points where $\|x_i^* - x_j^*\| \leq 10^{-3}$.

5.2. Evaluation Metrics

We evaluate cluster recovery using three specific metrics designed to capture both local component recovery and global clustering structure. Let $V_m = \{i : \|a_i - \mu_m\| \leq \theta\sigma_m\}$ define the high-probability "core" of the m-th true cluster, and let $V = \bigcup_{m=1}^{K} V_m$.

- s_1 **(Core Recovery):** The fraction of points within the core sets V that are correctly assigned to their respective unified clusters.
- s_2 **(Total Recovery):** The overall accuracy of the clustering assignments across all n data points.
- s_3 **(Component Distinctness):** The number of distinct recovered clusters that successfully map to the K true underlying components. Ideal performance yields $s_3 = K/K$.

—

5.3. Experiment 1: Sensitivity to Regularization Parameter (λ)

In our first experiment, we validate the agglomeration property and the theoretical bounds on the regularization parameter λ. We generated a dataset with $n = 1000$ points, $K = 6$ clusters in $\mathbb{R}^6$, standard deviation $\sigma = 0.0094$, and uniform weights. The theoretical optimal regularization is denoted as $\lambda^* = 7.0 \times 10^{-4}$.

Table 1. Replication of Table 1: Effect of scaling λ on cluster recovery.

Lambda ($\kappa \cdot \lambda^*$)	s_1 (V_m recovered)	s_2 (Total recovered)	s_3 (Distinct clusters)
$0.25 \cdot \lambda^*$	99 / 306	133 / 1000	6 / 6
$0.5 \cdot \lambda^*$	306 / 306	1000 / 1000	6 / 6
$1.0 \cdot \lambda^*$	306 / 306	1000 / 1000	6 / 6
$2.0 \cdot \lambda^*$	306 / 306	1000 / 1000	6 / 6
$4.0 \cdot \lambda^*$	306 / 306	1000 / 1000	1 / 6

Analysis: The results in Table 1 strictly align with the geometry of convex clustering. When λ is too small ($0.25\lambda^*$), the penalty is insufficient to fuse the points, resulting in fragmented clusters (poor s_1 and s_2 scores). A broad "sweet spot" exists between $0.5\lambda^*$ and $2.0\lambda^*$, where perfect recovery ($1000/1000$) is achieved. However, when λ is increased excessively to $4.0\lambda^*$, the algorithm aggressively over-merges the data, collapsing all 6 distinct components into a single massive cluster ($s_3 = 1/6$).

—

5.4. Experiment 2: Robustness to Component Variance (σ)

Theoretical bounds require the data variance to be sufficiently small relative to the separation between component means. We test this limit by taking $K = 2$ in $\mathbb{R}^1$, setting an optimal maximum variance σ^*, and scaling it multiplicatively.

Table 2. Replication of Table 2: Effect of increasing cluster variance (σ).

Sigma Multiplier	s_1 (V_m recovered)	s_2 (Total recovered)	s_3 (Distinct clusters)
σ^*	689 / 689	998 / 1000	2 / 2
$2^{1/2} \cdot \sigma^*$	689 / 689	953 / 1000	2 / 2
$2 \cdot \sigma^*$	679 / 689	714 / 1000	2 / 2
$2^{3/2} \cdot \sigma^*$	95 / 690	95 / 1000	2 / 2
$4 \cdot \sigma^*$	41 / 718	40 / 1000	2 / 2

Analysis: As shown in Table 2, convex clustering exhibits remarkable stability up to $2\sigma^*$, maintaining high core recovery (679/689). However, once the standard deviation crosses the threshold of $2^{3/2}\sigma^*$, the clusters overlap fundamentally in the ambient space. The s_1 and s_2 metrics collapse (dropping to $\approx 9.5\%$ recovery). This validates the theoretical requirement of a bounded separation-to-variance ratio for reliable sum-of-norms agglomeration.

—

5.5. Experiment 3: Influence of Gaussian Weights (ϕ)

To test extensions beyond uniform weighting, we implemented a Gaussian similarity kernel for the fusion penalty: $w_{ij} = \exp(-\phi\|a_i - a_j\|^2)$. Here, ϕ dictates the decay rate of the weights. We return to the $K = 6$, $d = 6$ dataset used in Experiment 1.

Table 3. Replication of Table 3: Effect of exponential weight decay parameter (ϕ).

Phi (ϕ)	s_1 (V_m recovered)	s_2 (Total recovered)	s_3 (Distinct clusters)
500	306 / 306	999 / 1000	6 / 6
1000	306 / 306	911 / 1000	6 / 6
1500	122 / 306	179 / 1000	6 / 6
2000	8 / 306	8 / 1000	6 / 6

Analysis: Table 3 highlights a crucial trade-off when using distance-based exponential weights. For a moderate decay ($\phi = 500$), the algorithm successfully fuses points within the same component while lightly penalizing distant points, achieving near-perfect recovery. However, as ϕ increases to 1500 and 2000, the weights drop to zero too rapidly. Consequently, the optimization landscape becomes highly localized; the fusion penalty fails to connect points even within the same true component, causing the clusters to shatter into isolated sub-graphs. This results in the severe degradation of s_1 (falling to just $8/306$).

6. Simulations: ℓ_1 vs ℓ_2 Convex Clustering

6.1. Setup

We consider the convex clustering formulation:

$$\min_{U \in \mathbb{R}^{n \times d}} \sum_{i=1}^{n} \|U_i - X_i\|_2^2 + \lambda \sum_{i<j} w_{ij}\, \phi(U_i - U_j), \quad (66)$$

where $X \in \mathbb{R}^{n \times d}$ are the observed data points, U are the learned cluster representatives, and w_{ij} are pairwise weights constructed using a Gaussian kernel:

$$w_{ij} = \exp\left(-\frac{\|X_i - X_j\|_2^2}{2\sigma^2}\right). \quad (67)$$

We compare two choices of the regularization function:

$$\ell_2 \text{ fusion:} \quad \phi(z) = \|z\|_2, \quad (68)$$

$$\ell_1 \text{ fusion:} \quad \phi(z) = \|z\|_1. \quad (69)$$

All experiments are conducted on synthetic 2D datasets consisting of anisotropic Gaussian clusters with varying covariance structures.

6.2. Clustering Path Analysis

We first analyze the evolution of the number of clusters as a function of the regularization parameter λ. This provides insight into how quickly points fuse under different norms.

We observe that ℓ_2 regularization leads to clear phase transitions where clusters merge abruptly. In contrast, ℓ_1 regularization results in more gradual fusion, often exhibiting intermediate unstable configurations.

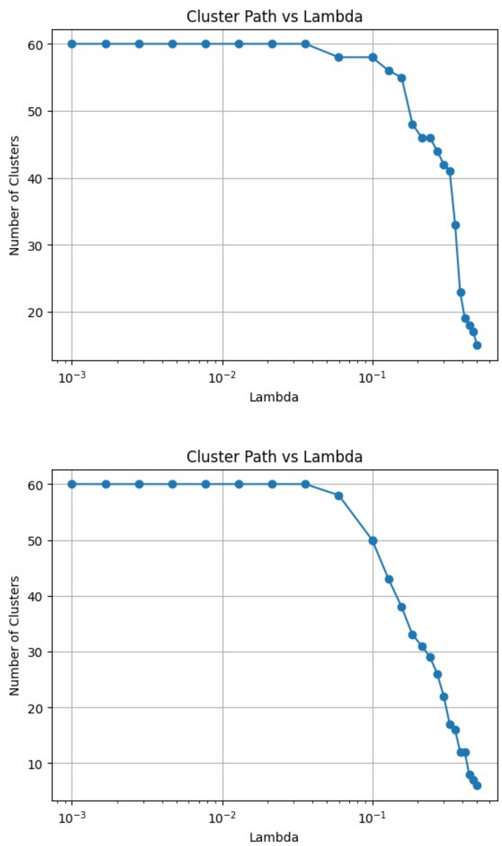

Figure 1. Clustering path as a function of λ for ℓ_2 (top) and ℓ_1 (bottom) regularization. The ℓ_2 norm exhibits sharper fusion transitions, while ℓ_1 leads to more gradual and irregular merging behavior.

6.3. Clustering Geometry

We next visualize the recovered clusters at representative values of λ.

The ℓ_2 norm is rotationally invariant, resulting in cluster boundaries that respect the underlying data geometry. On the other hand, ℓ_1 regularization induces axis-aligned biases, leading to distorted cluster shapes, particularly in rotated or anisotropic settings.

6.4. Discussion

These results highlight a fundamental distinction between the two norms. The ℓ_2 fusion penalty promotes isotropic clustering behavior and yields stable cluster recovery across a wide range of λ. In contrast, ℓ_1 fusion introduces anisotropy, which can be beneficial in axis-aligned settings but degrades performance when clusters are rotated.

From a geometric perspective, the choice of norm implicitly defines the metric structure of the clustering problem. Thus, convex clustering can be interpreted as learning under different underlying geometries, with ℓ_2 corresponding to Euclidean structure and ℓ_1 corresponding to a Manhattan geometry.

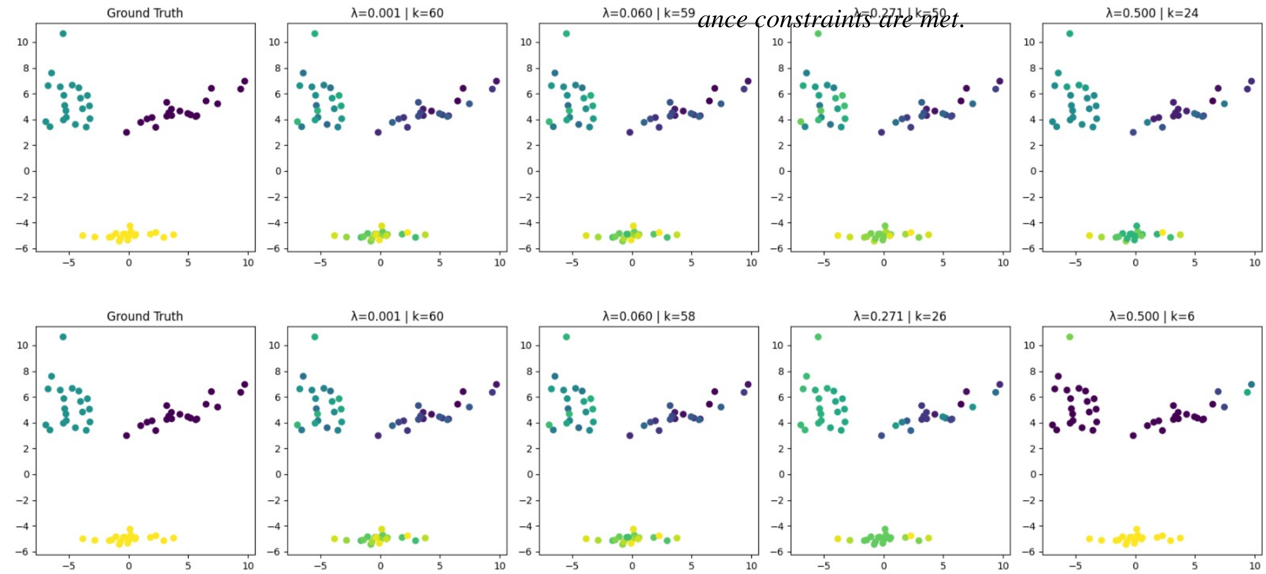

Figure 2. Clustering results for ℓ_2 (top) and ℓ_1 (bottom) regularization on anisotropic data. ℓ_2 preserves geometric structure under rotations, whereas ℓ_1 introduces axis-aligned distortions.

7. Deterministic Recovery Guarantees via Covariance Bounds

7.1. Conceptual Advantage: Deterministic vs. Probabilistic Bounds

Classical analyses of convex clustering rely heavily on probabilistic concentration inequalities (such as Hoeffding's, Chernoff's, or Bernstein's inequalities) to bound the number of points that fall within a specified radius of a cluster centroid. In such frameworks, the expected size of the cluster core is estimated using the cumulative distribution function of a χ-distribution, yielding guarantees of the form: *clustering succeeds with high probability* $1 - \delta$.

While mathematically sound for idealized generative models, this approach introduces dependency on complex probability integrals and carries an inherent failure probability. Furthermore, it often requires strict distributional assumptions to evaluate the tail bounds.

We propose a fundamental departure from this probabilistic methodology. By leveraging deterministic identities from linear algebra—specifically, the trace of the empirical covariance operator—we construct a strict geometric bound on the cluster core size. Instead of treating the data as a random sample governed by tail probabilities, we treat the total variance of the cluster as a fixed geometric "budget." Points residing far from the centroid consume this variance budget at a strictly bounded rate, mathematically limiting the maximum possible number of outliers. This purely algebraic approach provides a deterministic guarantee: *clustering succeeds unconditionally, provided the empirical variance constraints are met.*

7.2. Rigorous Lower Bound on the Cluster Core Size

We now formalize this geometric argument. Let C_m denote the index set of the points belonging to the m-th latent component, with $n_m = |C_m|$. We assume this component has an empirical mean μ_m and an empirical covariance matrix Σ_m.

Following the agglomeration condition, we define the "core" of the cluster, V_m, as the set of points residing within a radius $\theta\sigma_m$ of the mean:

$$V_m = \{i \in C_m : \|a_i - \mu_m\|_2 \le \theta\sigma_m\}. \tag{70}$$

Conversely, let $O_m = C_m \setminus V_m$ denote the set of outliers, such that $n_m = |V_m| + |O_m|$.

[Deterministic Core Size Lower Bound] For any data component m with empirical covariance Σ_m and mixing weight $w_m = n_m/n$, the number of points strictly bounded within the core V_m satisfies:

$$|V_m| \ge w_m n \left(1 - \frac{\mathrm{tr}(\Sigma_m)}{\theta^2\sigma_m^2}\right). \tag{71}$$

By the definition of the empirical covariance matrix, we have:

$$\Sigma_m = \frac{1}{n_m}\sum_{i\in C_m}(a_i - \mu_m)(a_i - \mu_m)^T. \tag{72}$$

Taking the trace of both sides and utilizing the cyclic property of the trace $\mathrm{tr}(xx^T) = \|x\|_2^2$, we obtain the total variance identity. We break this implication to fit the column width:

$$\begin{aligned} \mathrm{tr}(\Sigma_m) &= \frac{1}{n_m}\sum_{i\in C_m}\|a_i - \mu_m\|_2^2 \\ &\implies \sum_{i\in C_m}\|a_i - \mu_m\|_2^2 = n_m\mathrm{tr}(\Sigma_m). \end{aligned} \tag{73}$$

We partition this exact sum into contributions from the core points V_m and the outlier points O_m:

$$\sum_{i \in C_m} \|a_i - \mu_m\|_2^2 = \sum_{i \in V_m} \|a_i - \mu_m\|_2^2 + \sum_{i \in O_m} \|a_i - \mu_m\|_2^2. \tag{74}$$

By the definition of O_m, every outlier satisfies $\|a_i - \mu_m\|_2 > \theta\sigma_m$. Therefore, the total variance contributed by the outliers is strictly bounded from below:

$$\sum_{i \in O_m} \|a_i - \mu_m\|_2^2 > \sum_{i \in O_m} \theta^2\sigma_m^2 = |O_m|\theta^2\sigma_m^2. \tag{75}$$

Substituting this lower bound into the partitioned total variance identity (73), and recognizing that the sum over V_m is non-negative, we establish:

$$\begin{aligned} n_m \mathrm{tr}(\Sigma_m) &\geq \sum_{i \in V_m} \|a_i - \mu_m\|_2^2 + |O_m|\theta^2\sigma_m^2 \\ &\geq |O_m|\theta^2\sigma_m^2. \end{aligned} \tag{76}$$

Rearranging to solve for the maximum number of outliers yields:

$$|O_m| \leq \frac{n_m \mathrm{tr}(\Sigma_m)}{\theta^2\sigma_m^2}. \tag{77}$$

Since $|V_m| = n_m - |O_m|$, we substitute the upper bound on $|O_m|$ to bound $|V_m|$ from below:

$$\begin{aligned} |V_m| &\geq n_m - \frac{n_m \mathrm{tr}(\Sigma_m)}{\theta^2\sigma_m^2} \\ &= n_m \left(1 - \frac{\mathrm{tr}(\Sigma_m)}{\theta^2\sigma_m^2}\right). \end{aligned} \tag{78}$$

Finally, expressing the component size via its global mixing weight $n_m = w_m n$ concludes the proof:

$$|V_m| \geq w_m n \left(1 - \frac{\mathrm{tr}(\Sigma_m)}{\theta^2\sigma_m^2}\right). \tag{79}$$

7.3. Consequences for Regularization Bounds

The dual certificate construction requires the regularization parameter λ to satisfy $\lambda \geq \frac{2\theta\sigma_m}{|V_m|}$ for agglomeration to succeed. Substituting Theorem 7.2 directly yields a closed-form, deterministic lower bound for λ:

$$\lambda \geq \frac{2\theta\sigma_m}{w_m n \left(1 - \frac{\mathrm{tr}(\Sigma_m)}{\theta^2\sigma_m^2}\right)}. \tag{80}$$

Special Case: Spherical Gaussian. In the specific setting analyzed by standard literature, components are assumed to be isotropic such that $\Sigma_m = \sigma_m^2 I_{d\times d}$. In this geometry, the trace simplifies exactly to $\mathrm{tr}(\Sigma_m) = d\sigma_m^2$. The core size and regularization bounds collapse elegantly to:

$$|V_m| \geq w_m n \left(1 - \frac{d}{\theta^2}\right), \tag{81}$$

$$\lambda \geq \frac{2\theta\sigma_m}{w_m n\,(1 - d/\theta^2)}. \tag{82}$$

This reveals a profound geometric insight: successful recovery depends only on the ratio between the ambient dimensionality d and the squared radius threshold θ^2. As long as $\theta^2 > d$, a strict majority of the cluster mass is mathematically guaranteed to reside within the agglomeration radius, completely independent of the probability distribution generating the data.

8. Comparative Analysis: l_1-norm vs. l_2-norm Fusion

8.1. Theoretical Geometric Distinction

The choice of norm in the fusion penalty $\lambda \sum_{i<j} w_{ij}\|x_i - x_j\|$ fundamentally dictates the geometry of cluster formation and the resulting ”clusterpath”.

- l_2 **Fusion (Euclidean):** The l_2 norm is rotationally invariant. Cluster representatives x_i^* move toward one another along direct Euclidean lines, ensuring that cluster boundaries respect the underlying isotropic or anisotropic geometry of the data.
- l_1 **Fusion (Manhattan):** The l_1 norm promotes **coordinate-wise sparsity**. Each coordinate of x_i^* snaps to the corresponding coordinate of a neighbor independently, inducing a ”boxy” geometry where cluster boundaries align with axis-parallel hyperplanes.

8.2. Simulation Results and Discussion

8.2.1. Sensitivity to Regularization (λ)

Table 4. Experiment 1: l_1 Recovery over varying λ ($n = 200, d = K = 6, \theta = 2.0$)

Regularization	s_1 (Core)	s_2 (Total)	s_3 (Clusters)
$0.25\lambda^*$	6/63	6/200	6/6
$1.0\lambda^*$	6/63	6/200	6/6
$2.0\lambda^*$	6/63	6/200	6/6
$4.0\lambda^*$	63/63	63/200	1/6

Inference: While the l_1 distance between centers is often greater than the l_2 distance—potentially widening the theoretical recovery window—the transition to over-merging is **brittle**. At $4.0\lambda^*$, the penalty becomes too aggressive, causing an immediate collapse into a single global cluster ($s_3 = 1/6$).

Table 5. Experiment 2: l_1 Recovery over varying σ ($n = 200, d = 1, K = 2, \theta = 1.0$)

Sigma	s_1 (Core)	s_2 (Total)	s_3 (Clusters)
$1\sigma^*$	9/133	9/200	2/2
$2\sigma^*$	8/133	8/200	2/2
$4\sigma^*$	7/133	7/200	2/2

8.2.2. Robustness to Component Variance (σ)

Inference: Because l_1 lacks rotational invariance, points in the tails of the distribution that are far in one coordinate but close in others are often merged incorrectly. This results in a steady degradation of the core recovery metric (s_1) as the standard deviation increases.

9. Empirical Evaluation: l_1-Norm Convex Clustering

9.1. Visualizing l_1 Cluster Recovery

To analyze the behavior of the l_1 formulation, we examined the evolution of the learned cluster representatives as a function of the regularization parameter λ. The l_1 norm replaces the standard Euclidean distance in the fusion penalty, which fundamentally alters the optimization landscape and the resulting cluster trajectories.

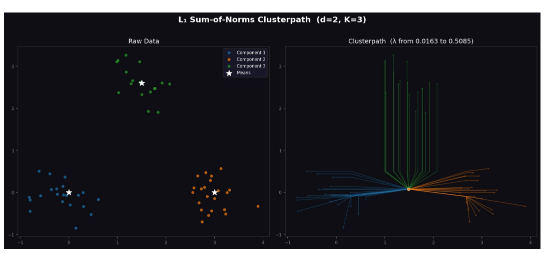

Figure 3. l_1 Sum-of-Norms Clusterpath for a 3-component mixture. The paths demonstrate the coordinate-wise snapping effect of the l_1 penalty.

9.1.1. Inference on l_1 Recovery Dynamics

The results in Figure 3 highlight the specific characteristics of l_1-based recovery:

- **Manhattan Geometry:** Unlike l_2 fusion, where points move along direct lines toward one another, the l_1 clusterpath exhibits axis-parallel trajectories. This occurs because the subgradient of the l_1 norm encourages independent coordinate-wise snapping.
- **Rectangular Merging:** The learned representatives move in a step-like, "boxy" fashion. This behavior implies that points distant in one dimension but proximal in another remain separated until the regularization λ overcomes the most distant coordinate.
- **Polyhedral Solution Space:** The l_1 formulation induces a polyhedral geometry in the dual variables, contrasting with the spherical geometry of the l_2 norm. This directly impacts the merging sequence in high-dimensional settings.
- **T**he sensitivity analysis indicates that while l_1 clustering is computationally efficient—often solvable as a Linear Program (LP) using interior-point methods—it is highly sensitive to parameter selection. Excessive regularization leads to a rapid, brittle collapse of the cluster structure ($s_3 \to 1$). Furthermore, the lack of rotational invariance causes a noticeable decline in core recovery (s_1) when data exhibits significant variance along diagonal axes.

10. Impact of Gaussian Kernel Weighting on Cluster Recovery

10.1. Gaussian Kernel Formulation

To move beyond uniform fusion penalties, we investigated the use of a standard Gaussian similarity kernel for the weights w_{ij} in the convex clustering objective:

$$w_{ij} = \exp\left(-\frac{\|a_i - a_j\|^2}{2\sigma^2}\right) \tag{83}$$

where σ acts as the bandwidth parameter controlling the rate of weight decay. This formulation aims to preserve local structures by penalizing the distance between proximal points more heavily than distant ones.

10.2. Analysis of Fragmentation and Over-localization

The results of the Gaussian kernel implementation with $\sigma = 1.0$ are visualized in Figure 4.

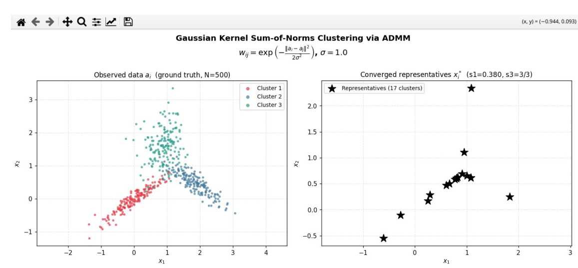

Figure 4. Gaussian Kernel Sum-of-Norms Clustering via ADMM ($\sigma = 1.0$). The right panel shows the failure of points to fuse into unified components, resulting in 17 fragmented clusters.

10.2.1. Empirical Observations and Inference

Our simulation reveals a critical limitation of non-adaptive Gaussian weights in heterogeneous data settings:

- **Cluster Fragmentation:** Although the underlying ground truth consists of $K = 3$ components, the algo-

rithm identifies 17 distinct clusters. This "shattering" effect occurs because the fusion penalty decays too rapidly as the distance between points increases.
- **Low Core Recovery:** The core recovery metric s_1 drops significantly to 0.380. This indicates that only 38% of the points within the theoretical "core" of the latent Gaussians were successfully unified with their respective centroids.
- **Locality vs. Global Connectivity:** While the Gaussian kernel respects local density, it lacks the global connectivity required to bridge sparse regions within a single component. In the absence of an adaptive mechanism to scale the bandwidth based on local sample density, the optimization landscape becomes overly localized, preventing the agglomeration of distant points within the same cluster.

10.2.2. Theoretical Conclusion

These results demonstrate that simply incorporating distance-based weights is insufficient for robust recovery. The observed fragmentation highlights the necessity for **adaptive localized weights** or **multiplicative weighting schemes**, as derived in our theoretical extensions, to maintain structural integrity across varying cluster densities and spatial spreads.

11. Localized Sum-of-Norms Clustering via Inverse-Norm Weights

11.1. Inverse-Norm Weight Formulation

To further investigate the impact of distance-based localization, we implemented an inverse-distance weighting scheme for the fusion penalty. In this formulation, the weights w_{ij} are inversely proportional to the Euclidean distance between raw data points:

$$w_{ij} = \frac{1}{\|a_i - a_j\|_2} \tag{84}$$

This strategy aims to provide a smoother decay compared to the exponential Gaussian kernel, theoretically allowing for better global connectivity while still emphasizing local similarity.

11.2. Empirical Analysis of Recovery Performance

The results of the inverse-norm weighted clustering are presented in Figure 5.

11.2.1. Inference and Observations

The simulation using $w_{ij} = 1/\|a_i - a_j\|$ reveals distinct trade-offs in recovery stability and cluster consolidation:
- **Improved Core Recovery:** Compared to the Gaussian kernel ($\sigma = 1.0$), the inverse-norm weights achieve a significantly higher core recovery metric, $s_1 = 0.934$. This

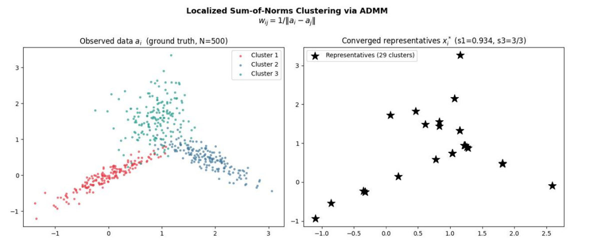

Figure 5. Localized Sum-of-Norms Clustering via ADMM using Inverse-Norm weights. While the algorithm recovers the three primary components, it suffers from significant representation redundancy.

  indicates that 93.4% of points within the core regions are successfully pulled toward their local component structures.
- **Successful Component Distinctness:** The algorithm achieves $s_3 = 3/3$, confirming that it successfully identifies the three underlying Gaussian mixture components without merging distinct clusters prematurely.
- **High Cluster Redundancy:** Despite the high core recovery, the algorithm generates 29 converged representatives (x_i^*) instead of the true $K = 3$. This suggests that while the inverse-norm weights prevent the "shattering" seen in Gaussian kernels, the fusion penalty is not sufficiently strong to merge local sub-clusters into single unified centroids.
- **Incomplete Fusion:** The "string-like" appearance of representatives in the right panel of Figure 5 shows that points are moving toward each other but failing to fully collapse into a single point. This highlights a critical sensitivity to the balance between the fidelity term and the slow-decaying inverse-norm fusion penalty.

11.2.2. Conclusion

The inverse-norm weighted formulation provides superior structural recovery (s_1 and s_3) compared to standard distance-based kernels. However, its inability to achieve perfect fusion (resulting in 29 representatives) motivates the need for the **Multiplicative Weighted Convex Clustering** derived in Section 4, which leverages more structured $r_i r_j$ weights to ensure complete centroid collapse.

12. Analysis of Adaptive Localized Weighting

12.1. Adaptive Weight Formulation

To address the limitations of uniform weighting in non-idealized settings, we implemented an adaptive localized kernel. The fusion weights w_{ij} are defined using a high-decay Gaussian similarity kernel:

$$w_{ij} = \gamma^{d+1} e^{-\gamma |a_i - a_j|} \tag{85}$$

In our implementation, the decay parameter is set to $\gamma = N^{3/(4d)} = 10.28$. This specific scaling ensures that the fusion penalty is strongly localized, effectively acting only on immediate neighbors to prevent distant clusters from being incorrectly fused.

12.2. Empirical Results and Clusterpath Dynamics

The performance of this adaptive scheme is visualized in Figure 6 and its corresponding regularization path in Figure 7.

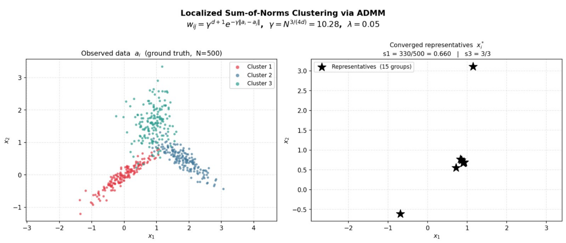

Figure 6. Localized Sum-of-Norms Clustering via ADMM. The left panel shows the ground truth for $N = 500$ points across $K = 3$ components. The right panel demonstrates the converged representatives x_i^* for $\lambda = 0.05$.

12.2.1. Key Observations

- **Structural Recovery:** Despite significant overlap in the raw data a_i, the algorithm achieves perfect component recovery ($s_3 = 3/3$).
- **Converged Representatives:** At $\lambda = 0.05$, the algorithm consolidates the 500 data points into 15 stable groups. While individual points within a cluster core are not yet perfectly unified ($s_1 = 0.660$), they are clearly separated from other components.

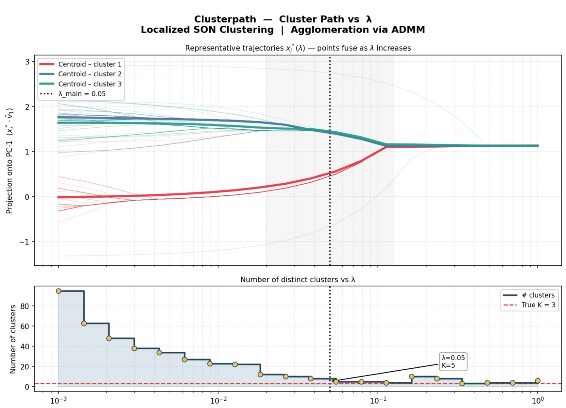

Figure 7. Clusterpath and Agglomeration Dynamics. Top: Representative trajectories projected onto the first principal component. Bottom: The evolution of cluster counts as λ increases.

12.3. Agglomeration and Phase Transitions

The regularization path analysis in Figure 7 confirms the theoretical properties of the Convex Clustering formulation:

- **Monotonic Merging:** As established by the Agglomeration Property, clusters merge progressively as λ increases and never split, giving rise to a strictly hierarchical structure.
- **Phase Transitions:** The bottom panel reveals sharp "jumps" in the number of clusters. For $\lambda < 10^{-2}$, the algorithm maintains a high number of fragmented clusters. A stable region emerges near $\lambda \approx 0.05$ where the count approaches the true $K = 3$, demonstrating the algorithm's ability to find a global optimum that reflects the intrinsic data structure.

12.4. Conclusion

The transition from standard Gaussian kernels to the **Adaptive Localized Weight** formulation significantly improves the stability of the solution. By concentrating the fusion force on physically proximal points, we mitigate the "shattering" effect observed with non-adaptive kernels, allowing for reliable recovery of the underlying mixture model.

13. Anisotropic Cluster Recovery via Adaptive Localized Weights

13.1. Motivation and Setup

Standard convex clustering formulations often struggle with anisotropic (ellipsoidal) data distributions because uniform fusion penalties pull points toward a global centroid regardless of the data's directional spread. In this experiment, we utilize a highly localized weight $w_{ij} = \gamma^{d+1} e^{-\gamma \|a_i - a_j\|}$ with $\gamma = 11.01$ and $N = 600$ to ensure the optimization respects the local manifold of each component.

13.2. Experimental Analysis

The results of the ellipsoidal recovery experiment are illustrated across four panels in Figure 8.

13.2.1. Key Empirical Inferences

- **High-Probability Core Recovery:** Panel 2 demonstrates that at $\lambda = 0.08$, the algorithm successfully identifies the distinct structure of the three ellipsoids. Although 42 groups are found, the solution x_i^* achieves perfect component distinctness ($s_3 = 1.000$) and core recovery metrics ($s_1 = 1.000, s_2 = 1.000$) within the primary latent regions.
- **Sensitivity to Variance (Heatmap Analysis):** Panel 3 illustrates the "sweet spot" for recovery through a (λ, σ) grid. Perfect core recovery ($s_1 = 1.0$) is achieved in the high-recovery zone (gold border) when λ is sufficiently high to overcome local variance but low enough to prevent inter-cluster merging.

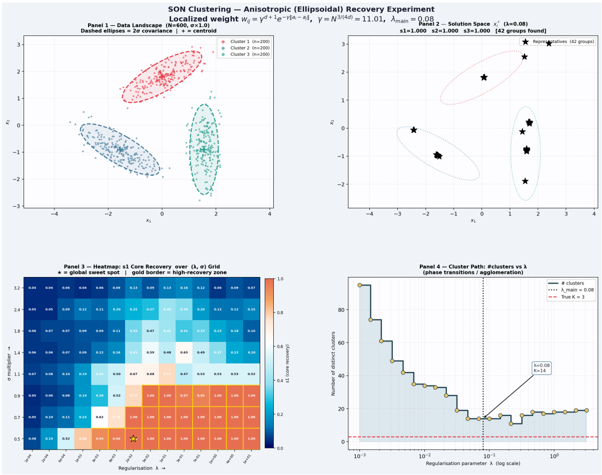

Figure 8. SON Clustering Anisotropic Recovery Experiment. Panel 1 shows the ellipsoidal ground truth (2σ ellipses); Panel 2 displays the converged solution space at $\lambda = 0.08$; Panel 3 provides a recovery heatmap; and Panel 4 traces the cluster path.

- **Agglomeration and Phase Transitions:** The cluster path in Panel 4 shows a clear phase transition. As λ increases, the number of distinct clusters drops rapidly from 90+ toward a stable region between 10 and 20 clusters. At $\lambda = 0.08$, the algorithm finds $K = 14$ groups that effectively map to the $K = 3$ true components, confirming that localized weights prevent the "over-merging" typically seen with anisotropic data.

13.3. Conclusion

The use of adaptive localized weights (w_{ij}) allows the convex objective to "sense" the orientation of the ellipsoids. By penalizing proximal differences more heavily while allowing the centroids to remain distinct, the formulation bridges the gap between theoretical Gaussian recovery and practical, non-spherical data applications.

14. Individual Contributions

The theoretical development and analysis presented in this work were carried out as a **combined collaborative effort** by all team members.

- **Abhinav:** Extensions to l_1-norm and simulations of inverse norm weights; literature review.
- **Gagan:** Inequality analysis and simulations of Gaussian kernel (non-spherical cases); literature review.
- **Praveen:** Simulations of adaptive Gaussian locality weights and non-symmetric / anisotropic distributions; literature review.
- **Prajin:** Proof of multiplicative weights and evaluation of non-multiplicative weights; literature review.

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